

ex.9

```
1 clear;
2 clc;
3 close all;
4
5 %% PARAMETERS
6
7 BW = 100e6; % Bandwidth = 100 MHz
8
9 SNR_dB = 0:5:30; % SNR range in dB
10
11 mod_schemes = {'QPSK','16QAM','64QAM'};
12
13 bits_per_symbol = [2 4 6]; % Bits per symbol
14
15 throughput = zeros(length(mod_schemes),length(SNR_dB));
16
17 %% CALCULATE THROUGHPUT
18
19 for m = 1:length(mod_schemes)
20
21     k = bits_per_symbol(m);
22
23     for s = 1:length(SNR_dB)
24
25         % Convert SNR from dB to linear
26         snr_linear = 10^(SNR_dB(s)/10);
27
28         % Shannon capacity
29         C = BW * log2(1 + snr_linear);
30
31         % Modulation efficiency adjustment
32         throughput(m,s) = C * (k/6);
33
34     end
35
36 end
```

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Run Save Output Input

SNR = 5 dB : 137.16 Mbps
SNR = 10 dB : 230.63 Mbps
SNR = 15 dB : 335.19 Mbps
SNR = 20 dB : 443.88 Mbps
SNR = 25 dB : 553.96 Mbps
SNR = 30 dB : 664.48 Mbps

64QAM Throughput:
SNR = 0 dB : 100.00 Mbps
SNR = 5 dB : 205.74 Mbps
SNR = 10 dB : 345.94 Mbps
SNR = 15 dB : 502.78 Mbps
SNR = 20 dB : 665.82 Mbps
SNR = 25 dB : 830.94 Mbps
SNR = 30 dB : 996.72 Mbps

[Execution complete with exit code 0]

